MANIPAL INSTITUTE OF TECHNOLOGY
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SEVENTH SEMESTER B.TECH. (CIVIL ENGINEERING)

END SEMESTER EXAMINATION, NOV 2022

COASTAL ENGINEERING (CIE 4076)

(21–11 - 2022)

TIME: 3 HRS. **MAX. MARKS: 50**

Note: 1. Answer all questions.
2. Any missing data may be suitably assumed.

Q. NO	QUESTION	MARKS	CO
1A	Determine the wave period and celerity for a wave having a length of 75 m in 8 m depth of water using Airy's wave theory.	**2**	**1**
1B	Specify the Boundary conditions used for obtaining the velocity potential 'Φ' of a linear water wave problem.	**3**	**1**
1C	Obtain dispersion relation using equation of velocity potential.	**5**	**1**
2A	With a neat sketch, explain how water particle displacements vary at various water depths.	**2**	**3**
2B	List out the natural and manmade causes for coastal erosion	**3**	**3**
2C	Plot the distribution of maximum horizontal and maximum vertical water particle velocities across depth at 2 m interval. The ocean wave measured at 10 m is having a wave height of 2.5 m and wave period of 8 sec.	**5**	**1**
3A	Determine if Sainflou's method can be adopted for a vertical wall breakwater design of 15 m height. The water depth at site is 10 m with a wave period of 10 s. The refraction diagram gave spacing between adjacent orthogonals in deep water to be 2 units and at structure site as 3 units. Consider shoaling coefficient of 0.9981 for a deepwater wave height of 5 m.	**2**	**3**
3B	With a neat sketch differentiate between graving and floating dry dock.	**3**	**4**
3C	What is a berthing structure? With the help of neat diagram explain various types of berthing structures.	**5**	**5**
4A	Compute H_{mean}, $H_{1/10}$, $H_{1/3}$ for the short term wave data recorded over 20 mins duration which yielded 100 waves.	**3**	**5**
4B	What are the requirements of a good harbor?	**3**	**4**
4C	What are breakwaters? Give a detailed classification of breakwaters with figures.	**4**	**5**
5A	Design and draw a cross-section of Trunk of rubble mound breakwater at a location with following details	**5**	**5**

H in m	0-0.5	0.6-1.1	1.2-1.7	1.8-2.3	2.4-2.9	3-3.5
No. of occurrences	21	25	20	18	10	6

	Mean water depth – 10 m Design wave height - 4m, Wave period-9 Sec Wave runup - 4 m Tidal correction – 1.5 m Armour unit – Quarried natural rocks up to 125 kN size with Unit weight of 2.6 t/m^3 can be used. Take K_D – 2.8 and K_Δ - 1.		
5B	Write a short note on coastal eco-systems	**3**	**5**
5C	What is Coastal pollution and its implications?	**2**	**5**